Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/28/2007
PI Name:	Wu, Xiao-lun Ph.D.		
Project Title:	Full-field Interferometric Measurements of Thickness of Freely-suspended Liquid Films		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid transport phenomena		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
Space Biology Element:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
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	We propose to develop interferometric methods to visualize and measure the film thickness in a two-dimensional flowing soap film. Since the film used can be sometimes several optical wavelengths in thickness, enumeration of the interference fringes resulting from a single wavelength in an interferogram is ambiguous and cannot be used to determine the precise value of the thickness. This difficulty can be overcome if interferograms are acquired simultaneously using several different wavelengths or in combination with infrared absorption measurements. The full-field thickness information on flowing liquid films is useful for understanding stress distribution in the film and how this stress relaxes over time and distance. The technique is also useful for understanding the interaction between turbulence and thickness fluctuations, which under certain conditions can be treated as a passive scalar similar to a dye or a temperature distribution.		

Task Description:	Build a flowing soap film channel that allows independent control of the film thickness and flow speed. Simple fluid flow structures, such as a single vortex or a few vortices created by a bluff body will be examined first. More complicated structures, such as a large array of vortices will be studied afterwards. Until recently most of studies of two-dimensional turbulence in flowing soap films have been concentrated on velocity fluctuations. However, our experience shows that accompanying the velocity fluctuations, the thickness of the film also fluctuates. The dynamic role of the thickness variation is not clear at this point. Theoretically, it is predicted that for the typical velocity of the flow to be least than the sound velocity in the film, the thickness variation is passively convected. On the other hand, when the typical velocity of the flow is greater than the sound velocity, the shock waves are created and the thickness variation is no longer passive. The two-dimensional experiment proposed here allows us to investigate the intricate coupling between the velocity field and the flow medium. Similar problems occur naturally in oceans and in atmosphere.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We have shown the importance of long chain polymers in the suppression of fluid flows and turbulence. The effect has been previous studied but is not well understood. Our finding points to the significance of hydrodynamic structures, particularly the saddle points in the flow, that are responsible for suppression of turbulence. The finding contributes to our general understanding of polymer effects in flows and may be useful for enhancing fluid transport in pipes and navagation efficiency. The finding that single-point velocity statistics is non-gaussian in 2D turbulence is also intriguing and will motivate new investigations in 3D turbulence.
Task Progress:	Project Progress 2005 and 2006 We have made much progress in the study of turbulence in flowing soap films that are laboratory realizations of two-dimensional fluid flows and turbulence. I would consider the most significant progress over the last two years to be in the areas of education and training students at all levels. Thanks to the NASA support, two students (Jie Zhang and Yonggun Jun) have successfully defended their Ph.D. theses and are now postdoctoral fellows at Duke University and at the Weizmann Institute of Science, respectively. Their current research projects are intimately related to their Ph.D. training. For instance Jie is working with a prominent physicist Robert Behringer on a project attempting to understand surface-tension driven fluid flows (to be discussed more below) and Yonggun is working in Victor Steinberg's lab investigating turbulence in a strongly viscoelastic fluid. It is of great personal satisfaction to see these young students gradually mature, becoming competent scientists who are not only well quipped with technical know-how but also with a positive outlook for life and career. I have also trained a superb undergraduate student, Pedram Roushan, who is now a physics graduate student at Princeton. Roushan worked in my laboratory for 3 years with significant achievements both in research and in his academic work. He received numerous awards both inside our university and nationwide. The most prestigious one was the Goldwater Scholarship, which supported his study at the University of Pittsburgh in the years of 2004 and 2005. As an educator and a scientist, training students is one of the most important jobs for me and I have taken this responsibility seriously. To continue my research in fluid turbulence, since we still have some funds left, I have recently recruited two more graduate students, Ildoo Kim and Kuwen Lee. So far they have made steady progress with research and I very hopeful that they will be as successful as my previous students. I have also recruited a minority student Lashanda Willis. Lashanda is in a work-study program supported by our university. In terms of research, we have concentrated on two main subjects (i) instrumental development that allows us to rapidly measure thickness fluctuations in fast moving liquid films and (ii) exploring fluid flow and turbulence using these very thin films, which are laboratory realizations of two-dimensional (2D) fluid media. In the following these two aspects of research are summarized. (i) Instrumental development. One of the aims of this research is to develop new techniques that allow researchers to measure the thickness of rapidly flowing liquid films. The thickness of the film is an important dynamic quantity because it appears explicitly in the equation of motion, and the product of the film thickness h and the density of water d yields the 2D mass density of the fluid, i.e. d_2=hd. The quantity contributes to the inertia of the motion and the kinematic viscosity of the 2D fluid. Reliable and rapid film thickness measurement in a freely suspended flowing film is challenging because of the demand of noninvasiveness, the speed, and a high signal/noise ratio in the measurement. We have implemented a novel infrared (IR) technique to measure film thickness using the state-of-the-art IR detector and a phase-lock-in method. The technique exploits the fact that long wavelength photons can readily escape from the interior of the film and can be detected by a sensitive IR detector. These thermalized photons carry useful information about the local environment of the fluid, such as temperature T and the density d. Indeed our calculation showed that radiation power emitted from a small patch of film is proportional to T^4, the typical contribution from a blackbody, and a prefactor, also known as the thermal emissivity epsilon that is a function of the film thickness. In the limit of a thin film, h<<lambda, we found that epsilon is linearly proportional to h and thus provides a convenient means that the important quantity d_2 can measured, where 8 um < lambda < 14 um is the IR wavelength by which the thermal radiation is detected. The IR technique also has the advantage that it is non-invasive and can be measured rapidly. My former graduate student Jie Zhang had spent considerable time designing, building, and calibrating this homebuilt IR radiation thermometer. The technique was the main tool he used for his thesis work and resulted in two publications in peer-reviewed journals [1,2]. The technical was also compared with results obtained from a commercial IR camera (a demonstration unit). The commercial camera though easier to use but it was considerably slower than our single-point measurement and costs more than $200k if purchased. This latter work is summarized in a short manuscript by Zhang and we wish to publish it shortly [3]. (ii) A great deal of physics has been done over the last two years. (a) We have investigated the intriguing phenomenon of large scale circulations (LSC) that are often observed in strongly convective region when a fluid is subject to a large temperature gradient [1]. In our 2D film, since we could follow both the velocity field v(x,y) and the density field d_2(x,y), we found that LSC occurs due to a density inversion. For a low temperature gradient, the film is stably stratified, i.e., it is thick in the bottom and is thin on the top. However, for the temperature gradient exceeds a critical value, the film is unstably stratified. The density inversion is intimately connected with the kinetic energy of thermal plumes that carries both heat and fluid mass. Above the threshold temperature gradient, the plumes are highly energetic and are able to rise to the top of the convection cell causing a significant increase in mass density on the top of the cell. We believe that similar mechanism may take place in 3D thermal convection but it remains

to be verified by further measurements. (b) We have determined that thermal convection in a 2D fluid is the Bolgiano type when the overall kinetic energy of thermal plumes is less than their corresponding potential energy. The measured 2D density spectrum is in rather good agreement with the classical predication using a Bolgiano's scaling relation [2]. When a very large temperature gradient is applied, the LSC occurs and the density power spectrum deviates strongly from the Bolgiano's prediction. However, surprisingly, in this strong convection regime, the 2D density behaves somewhat like a passive scalar and the spectrum can be explained by a Batchelor's scaling relation for dyes in turbulence [2]. (c) In a careful velocity measurement using particle imaging velocimetry, we discovered that the traditional view that velocity fluctuations in a single point in a turbulent flow obey Gaussian statistics is in fact incorrect [4]. Our conclusion was derived from a number of observations in both freely decaying and forced 2D turbulence. We were pleased to know that such a deviation was calculated using sophisticated field theoretical techniques by a group of theorists. According to this remarkable theory, the deviation is a general property of turbulence independent of spatial dimensions. However, ours was the first to show such an effect. The implication of this finding is broad and we believe many other stochastic processes, other than fluid turbulence, will display the same non-Gaussian behavior if the statistics is examined closed to the wings of the probability density functions. (c) We found that 2D turbulence is strongly intermittent when a thermal gradient is present [5]. This is again in sharp contrast with the common belief that 2D turbulence should be free of intermittency due to the lack of vorticity amplification in 2D space. We identified the source of intermittency to be due to a constant exchange of turbulent kinetic energy and potential energy of thermal plumes. Using lognormal statistics, Jie Zhang was able to calculate the intermittent index explicitly. The published work received excellent reviews after its submission.

Much is waiting to be discovered in this wonderful area of 2D turbulence research. Aside from further our fundamental understanding of this fascinating natural phenomenon, the work is also practically useful because fluid flows are common place and are important for many engineering problems and for weather prediction. We are grateful to the NASA grant that makes this work possible.

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[4] Single-Point Velocity Statistics of Forced and Decaying Two-Dimension Turbulence, Yonggun Jun, Jie Zhang, and X.L. Wu, Phys. Rev. Lett. 96, 164502 (2006).

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Bibliography Type:	Description: (Last Updated: 05/08/2019)
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